The empirical recurrence for OEIS A299011, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A299011 records "Empirical recurrence of order 66" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 71$. Its last coefficient is $c_{66} = 48 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299011 is "Number of nX4 0..1 arrays with every element equal to 0, 1, 2, 4, 5 or 6 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$8, 29, 44, 174, 1052, 4488, 18758, 89713, \ \dots$$

The entry states:

> Empirical recurrence of order 66 (see link above)

As of the "Last modified" line on the live entry (Jan 31 2018 09:36:18, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 5$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{66} c_i \, a(n-i),$$

c_1	6	c_{18}	-2757688	c_{35}	49861769	c_{52}	10400643
c_2	-10	c_{19}	3657929	c_{36}	-44278150	c_{53}	-4200379
c_3	56	c_{20}	-4614969	c_{37}	60160552	c_{54}	-5877171
c_4	-225	c_{21}	4784216	c_{38}	98394695	c_{55}	-4005537
c_5	260	c_{22}	-4110505	c_{39}	-2179457	c_{56}	-1850029
c_6	-900	c_{23}	1166583	c_{40}	110948781	c_{57}	-534171
c_7	3125	c_{24}	3396174	c_{41}	-107047489	c_{58}	5844
c_8	-5262	c_{25}	-5187070	c_{42}	67727720	c_{59}	513054
c_9	12255	c_{26}	20037800	c_{43}	-126942182	c_{60}	169543
c_{10}	-26102	c_{27}	-18465081	c_{44}	-22858139	c_{61}	56458
c_{11}	55166	c_{28}	22930658	c_{45}	-57110198	c_{62}	-22728
c_{12}	-121681	c_{29}	-25339086	c_{46}	6199946	c_{63}	-6200
c_{13}	229409	c_{30}	1251775	c_{47}	21383770	c_{64}	-1380
c_{14}	-389287	c_{31}	-8916164	c_{48}	-15578013	c_{65}	-272
c_{15}	680344	c_{32}	-42288840	c_{49}	3164114	c_{66}	48
c_{16}	-1253781	c_{33}	40887786	c_{50}	13941817		
c_{17}	1903288	c_{34}	-70245177	c_{51}	29238260		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 71$, and it is the recurrence OEIS A299011 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1 x^{66-1} - \cdots - c_{66}$. Its constant term is $-c_{66} = -48$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 48.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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